Exploring System Design of Netflix Part-4

Day 125
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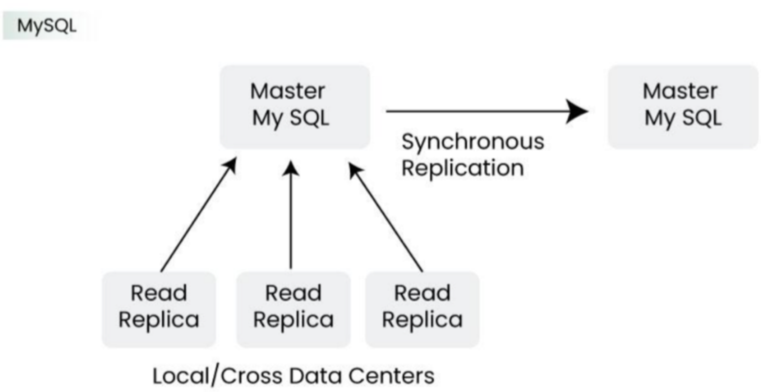
Data Base Design of Netflix

Netflix uses two different databases ie MySQL (RDBMS) and Cassandra (NoSQL) for different Purposes.

6.1. EC2 Deployed MySQL

> *Netflix saves data like billing information, user information, and transaction information in MySQL because it needs ACID compliance. Netflix has a master-master setup for MySQL and it is deployed on Amazon's large EC2 instances using InnoDB.*

The setup follows the "**Synchronous replication protocol**" where if the writer happens to be the primary master node then it will be also replicated to another master node. The acknowledgment will be sent only if both the primary and remote master nodes' write have been confirmed. This ensures the high availability of data. Netflix has set up the read replica for each and every node (local, as well as cross-region). This ensures high availability and scalability.



All the read queries are redirected to the read replicas and only the write queries are redirected to the master nodes.

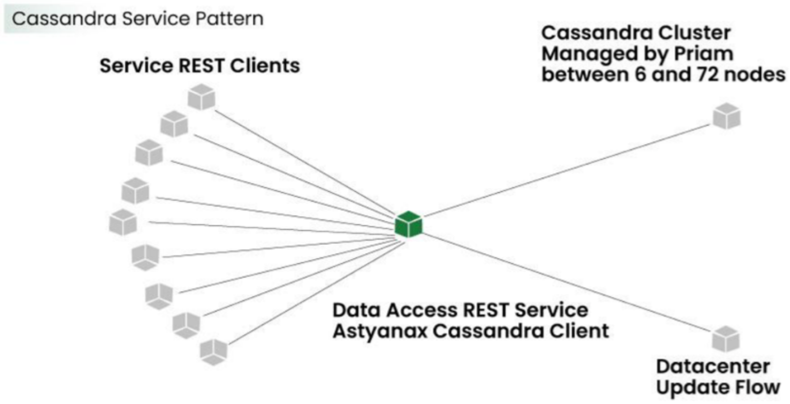
- In the case of a primary master MySQL failure, the secondary master node will take over the primary role, and the route53 (DNS configuration) entry for the database will be changed to this new primary node.
- This will also redirect the write queries to this new primary master node.

6.2. Cassandra

Cassandra is a NoSQL database that can handle large amounts of data and it can also handle heavy writing and reading. When Netflix started acquiring more users, the viewing history data for each member also started increasing. This increases the total number of viewing history data and it becomes challenging for Netflix to handle this massive amount of data.

Netflix scaled the storage of viewing history data-keeping two main goals in their mind:

- Smaller Storage Footprint.
- Consistent Read/Write Performance as viewing per member grows (viewing history data write-to-read ratio is about 9:1 in Cassandra).



Total Denormalized Data Model

- Over 50 Cassandra Clusters
- Over 500 Nodes
- Over 30TB of daily backups
- The biggest cluster has 72 nodes.
- 1 cluster over 250K writes/s

Initially, the viewing history was stored in Cassandra in a single row. When the number of users started increasing on Netflix the row sizes as well as the overall data size increased. This resulted in high storage, more operational cost, and slow performance of the application. The solution to this problem was to compress the old rows.

Netflix divided the data into two parts:

- **Live Viewing History (LiveVH):**
  Stores recent viewing data with frequent updates, kept uncompressed for fast ETL processing.
- **Compressed Viewing History (CompressedVH):**
  Stores older viewing records with rare updates, compressed to save storage space.

High Availability through Read Replicas